

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
MID SEM-1 EXAM
OPERATING SYSTEMS: (ITPC 200 Operating Systems/AIPC 202 Operating Systems/B.Tech (Mathematics & Computing: Operating Systems))

Max. Marks- 15 (Attempt any three questions. Each question carries 5 marks each)

Ques-1 In a real computer system, neither the resources available nor the demands of processes for resources are consistent over long periods (months). Resources break or are replaced, new processes come and go, new resources are brought and added to the system. If deadlock is controlled by the banker's algorithm, which of the following changes can be made safely (without introducing possibility of deadlock), and under what circumstances? (5 marks)

- a. Increase Available (new resources added)
- b. Decrease Available (resource permanently removed from system)
- c. Increase Max for one process (the process needs more resources than allowed, it may want more)
- d. Decrease Max for one process (the process decides it does not need that many resources)
- e. Increase the number of processes
- f. Decrease the number of processes

Ques-2(a) Consider the following preemptive priority-scheduling algorithm based on dynamically changing priorities. Larger priority numbers imply higher priority. When a process is waiting for the CPU (in the ready queue, but not running), its priority changes at a rate α ; when it is running, its priority changes at a rate β . All processes are given a priority of 0 when they enter the ready queue. The parameters α and β can be set to give many different scheduling algorithms. (2 marks)

(a) What is the algorithm that results from $\beta > \alpha > 0$?

(b) What is the algorithm that results from $\alpha < \beta < 0$?

Ques-2(b) Define a Process Control Block (PCB). List any four fields that are typically stored in a PCB. What is Context Switching? Why is it Overhead? (3 marks)

Ques-3 Consider a system consisting of m resources of the same type being shared by n processes. A process can request or release only one resource at a time. Show that the system is deadlock free if the following two conditions hold: (5 marks)

- a. The maximum need of each process is between one resource and m resources.
- b. The sum of all maximum needs is less than $m + n$.

Ques-4 Consider the following resource-allocation policy. Requests for and releases of resources are allowed at any time. If a request for resources cannot be satisfied because the resources are not available, then we check any processes that are blocked waiting for resources. If a blocked process has the desired resources, then these resources are taken away from it and are given to the requesting process. The vector of resources for which the blocked process is waiting is increased to include the resources that were taken away. For example, consider a system with three resource types and the vector Available initialized to (4,2,2). If process P_0 asks for (2,2,1), it gets them. If P_1 asks for (1,0,1), it gets them. Then, if P_0 asks for (0,0,1), it is blocked (resource not available). If P_2 now asks for (2,0,0), it gets the available one (1,0,0) and one that was allocated to P_0 (since P_0 is blocked). P_0 's Allocation vector goes down to (1,2,1), and its Need vector goes up to (1,0,1). (5 marks)

- a. Can deadlock occur? If you answer "yes," give an example. If you answer "no," specify which necessary condition cannot occur.
- b. Can indefinite blocking occur? Explain your answer.